Nama: Joaquin Thiogo

NPM : 240712831

Kelas : B

1. Activity 3

Contoh Jawaban Tabel pada IP 192.168.16.0/24:

Subnet (Jumlah Host)	Network Address	Subnet Mask	Host Range	Broadcast Address	Prefix
Tower B **(45 Hosts)**	192.168.16.0	255.255.255.192	192.168.16.1 – 192.168.16.62	192.168.16.63	/26
Tower A **(13 Hosts)**	192.168.16.64	255.255.255.240	192.168.16.65 – 192.168.16.78	192.168.16.79	/28
Router1-Router2 **(2 Hosts)**	192.168.16.80	255.255.255.252	192.168.16.81 – 192.168.16.82	192.168.16.83	/30

Cukup tulis rumus **2^n-2 ≥ Host yang dibutuhkan** di masing-masing subnet, tidak perlu tulis step yang lain.

Contoh:

1. **Tower B (45 Hosts)**
   $2^n-2 \geq 45$
   $2^6-2 \geq 45$
   $64 \geq 47$
   Prefix = 32 – n -> 32-6 = /26
2. **Tower A (13 hosts)**
   ....
3. **Router1-Router2 (2 hosts)**
   ....

For Your Information: Shortcut untuk buat 2^n di word itu, setelah 2 teman-teman bisa **Ctrl Shift +** untuk membuat pangkat. Kemudian untuk ≥ teman-teman bisa ketik 2265 di word lalu 2265nya teman-teman **Alt X** agar menjadi ≥. 😊

Step by step pengerjaan:

- IP Address

  240712831 = 2+4+0+7+1+2+8+3+1 = 28, maka IP Addressnya 10.28.32.19/8.

  SM = 255.0.0.0

  NA = 10.0.0.0/8

- Urutan
  1. Gedung B (61982 hosts)
  2. Gedung F (28912 hosts)
  3. Gedung E (10231 hosts)
  4. Gedung C (5671 hosts)
  5. Gedung D (3203 hosts)
  6. Gedung A (1234 hosts)
  7. WAN1: Router1-Router2 (2 hosts)
  8. WAN2: Router2-Router3 (2 hosts)
  9. WAN3: Router3-Router1 (2 hosts)

- Menentukan prefix
  1. **Gedung B (61982 hosts)**

     2^n-2 ≥ 61982

     2^{16}-2 ≥ 61982

     65536-2 ≥ 61982

     65534 ≥ 61982

     Prefix = 32 – n -> 32 - 16 = /16
  2. **Gedung F (28912 hosts)**

     2^n-2 ≥ 28912

     2^{15}-2 ≥ 28912

     32768-2 ≥ 28912

     32766 ≥ 28912

     Prefix = 32 – n -> 32 - 15 = /17
  3. **Gedung E (10231 hosts)**

     2^n-2 ≥ 10231

     2^{14}-2 ≥ 10231

     16384-2 ≥ 10231

     16382 ≥ 10231

     Prefix = 32 – n -> 32 - 14 = /18

4. **Gedung C (5671 hosts)**

   2^n-2 ≥ 5671

   2^{13}-2 ≥ 5671

   8192-2 ≥ 5671

   8190 ≥ 5671

   Prefix = 32 – n -> 32 - 13 = /19
5. **Gedung D (3203 hosts)**

   2^n-2 ≥ 3203

   2^{12}-2 ≥ 3203

   4096-2 ≥ 3203

   4094 ≥ 3203

   Prefix = 32 – n -> 32 - 12 = /20
6. **Gedung A (1234 hosts)**

   2^n-2 ≥ 1234

   2^{11}-2 ≥ 1234

   2048-2 ≥ 1234

   2046 ≥ 1234

   Prefix = 32 – n -> 32 - 11 = /21
7. **WAN1: Router1-Router2 (2 hosts)**

   2^n-2 ≥ 2

   2^2-2 ≥ 2

   4-2 ≥ 2

   2 ≥ 2

   Prefix = 32 – n -> 32 - 2 = /30
8. **WAN2: Router2-Router3 (2 hosts)**

   2^n-2 ≥ 2

   2^2-2 ≥ 2

   4-2 ≥ 2

   2 ≥ 2

   Prefix = 32 – n -> 32 - 2 = /30
9. **WAN3: Router3-Router1 (2 hosts)**

   2^n-2 ≥ 2

   2^2-2 ≥ 2

   4-2 ≥ 2

   2 ≥ 2

   Prefix = 32 – n -> 32 - 2 = /30

Jangan lupa Subnet di tabel diurutkan dari host **terbanyak** hingga host **tersedikit**!

Notes: Untuk urutan subnet router dibebaskan mau yang mana terlebih dahulu (misalnya Subnet Router1–Router2, Router2-Router3, Router3-Router1 atau mau Router2-Router3, Router3-Router1, Router1-Router2, dll) dengan catatan IP Addressnya masih **sesuai** dengan perhitungan VLSM.

Subnet (Jumlah Host)	Network Address	Subnet Mask	Host Range	Broadcast Address	Prefix
Gedung B **(61982 hosts)**	10.0.0.0	255.255.0.0	10.0.0.1 – 10.0.255.254	10.0.255.255	/16
Gedung F **(28912 hosts)**	10.1.0.0	255.255.128.0	10.1.0.1 – 10.1.127.254	10.1.127.255	/17
Gedung E **(10231 hosts)**	10.1.128.0	255.255.192.0	10.1.128.1 – 10.1.191.254	10.1.191.255	/18
Gedung C **(5671 hosts)**	10.1.192.0	255.255.224.0	10.1.192.1 – 10.1.223.254	10.1.223.255	/19
Gedung D **(3203 hosts)**	10.1.224.0	255.255.240.0	10.1.224.1 – 10.1.239.254	10.1.239.255	/20
Gedung A **(1234 hosts)**	10.1.240.0	255.255.248.0	10.1.240.1 – 10.1.247.254	10.1.247.255	/21
Router1-Router2 **(2 hosts)**	10.1.248.0	255.255.255.252	10.1.248.1 - 10.1.248.2	10.1.248.3	/30
Router2-Router3 **(2 hosts)**	10.1.248.4	255.255.255.252	10.1.248.5 - 10.1.248.6	10.1.248.7	/30
Router3-Router1 **(2 hosts)**	10.1.248.8	255.255.255.252	10.1.248.9 - 10.1.248.10	10.1.248.11	/30

2. Activity 4

Tidak perlu screenshot, cukup file **.pkt** saja.

3. Format Pengumpulan:

- **Activity 3: Modul9_X_YYYYY_Act3.pdf**
- **Activity 4: Modul9_X_YYYYY_Act4.pkt**
- **Dizip dalam 1 folder Modul9_X_YYYYY_2.zip**

X = Kelas Jarkom Praktikan, YYYYY = 5 digit terakhir NPM Praktikan